

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

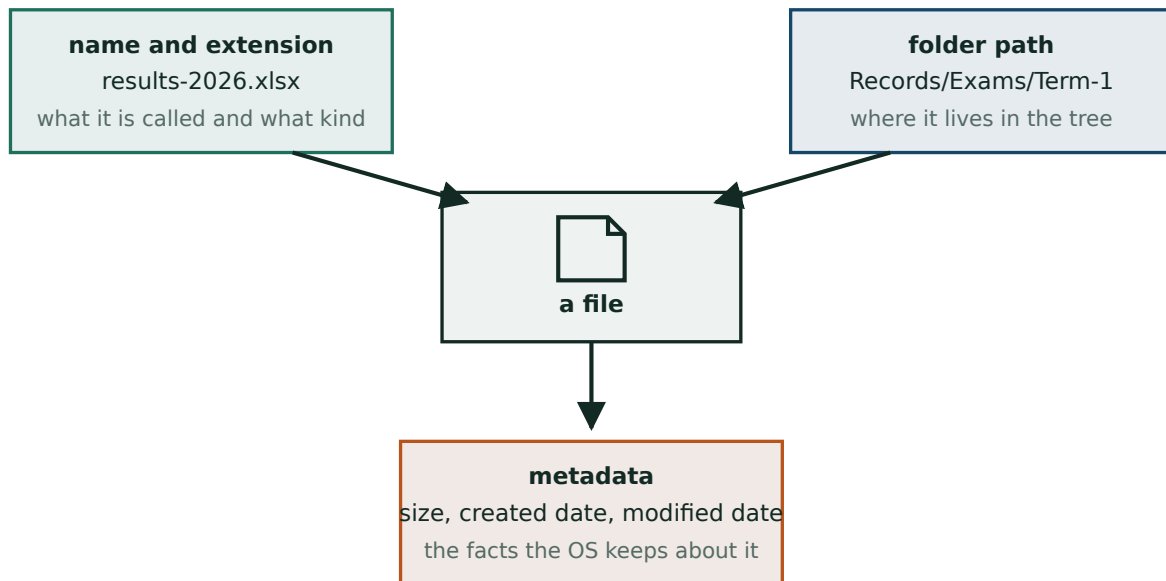
Topic 3.2 - Files and processes

A real classroom situation

Two scenes from the same staffroom. In the first, the clerk searches for "that letter" among four hundred files named "doc1", "doc2", "new final", and "new final FINAL" - the digital equivalent of a cupboard with no shelves. In the second, a trainee teacher complains that the computer "froze" when she opened her lesson file, the browser and a video call together; the clerk glances at the machine, closes one program, and everything moves again. Both scenes are about the same two jobs the operating system does for us all day: keeping files findable, and sharing the processor among running programs. The first scene is file management gone wrong. The second is process management at its limit.

What makes a file a file

Three identifying features, kept by the operating system



Deleting a file removes its record from the index; the space is marked reusable.

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Explanation

File management

A file is a named collection of data stored by the operating system - a document, a photo, a spreadsheet, a program. Three things identify a file. Its **name and extension** tell you what it is called and usually what kind of thing it is: "results-2026.xlsx" is a spreadsheet, ".jpg" an image, ".pdf" a document ready to print. Its **folder path** tells you where it lives: in "Records", inside "Exams", inside "Term-1". Folders (also called directories) nest inside one another, which is what turns a flat pile into a tree you can climb. And its **metadata** is the information the operating system keeps about the file: its size, when it was created, when it was last changed, who may open it.

The operating system's file management is the set of services built on those three things: creating, naming, renaming, moving, copying and deleting files; keeping the folder tree; and tracking where each file's actual bytes sit on the disk. When you delete a file, the OS typically removes its record from the index and marks its space as reusable - which is why a deleted file sometimes can be recovered, and why "deleted" is not the same as "erased".

The clerk in the first scene does not have a file problem; he has a naming and folder problem. A workable school scheme is three levels deep at most (year, then kind, then file), a date-first naming pattern ("2026-03-15-term1-results"), and no file sitting loose on the desktop. The scheme is not decoration: it is the difference between finding a record in five seconds and reconstructing a term's paperwork from memory.

Process management

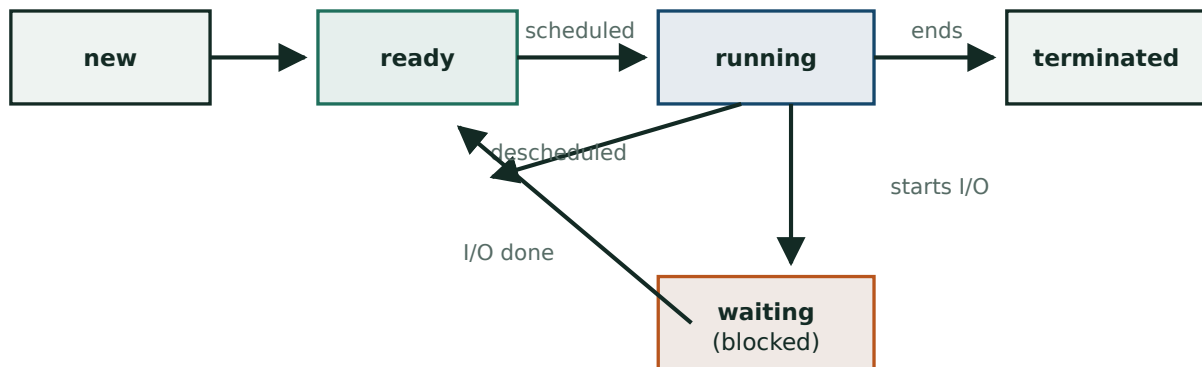
A process is a program in execution - a running program. Arpaci-Dusseau and Arpaci-Dusseau (2023) describe what makes one up: its address space, the memory it can use, holding its instructions, its data, and its stack; its registers, including the program counter that says which instruction comes next; and the list of files it currently has open.

Here is the part that changes how you see a "frozen" computer. One processor can run only one process at an instant, yet dozens appear to run together. The operating system creates that illusion by time sharing: run one process briefly, stop it, run another, and switch so fast that every program seems continuous. Arpaci-Dusseau and Arpaci-Dusseau (2023) give the simplified three states of a process. **Running**: the process is executing instructions on the processor.

Ready: the process could run, but the OS has chosen someone else for the moment. **Blocked**: the process has started an operation - say, reading from a disk - and cannot usefully continue until it finishes. A process moves from ready to running when it is scheduled, back from running to ready when it is descheduled, and into blocked when it must wait; when its work ends, it terminates. Add the beginning (new) and the end (terminated) and you have the process lifecycle.

The life of a process

One processor, rapid turns: the OS moves every program through these states



Too many processes wanting turns means a smaller slice for each: the slow machine. textbook.com.pk

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Now the trainee's "frozen" computer reads differently. Too many processes want turns; each gets a smaller slice; everything crawls. The fix is not mysterious: close processes, and the remaining ones get bigger slices. When a program's window closes but its process stays alive in the background (which happens), the machine still feels the load - and the operating system's process list is where you can see and end it.

Activity: The naming rescue and the process list

Work in pairs. About 20 minutes.

1. Your teacher gives you a list of fifteen badly named files in a jumble ("doc1", "untitled", "final2", "scan001", "asdf") with their real contents described (term results, a parent notice, a photo of the science fair, and so on).
2. Design a folder scheme (at most three levels) and rename every file with a date-first pattern. Write the full path of five of them.
3. Then open the running-programs or task view on the classroom machine (or your phone's app switcher). List what you find, mark which are visible

applications and which are background processes, and identify one you could safely close on a slow machine.

4. Swap with another pair: can they find any file in your scheme in under ten seconds just from its description?

Check your understanding

1. Define a file and a process in one sentence each. (*Remember*)
2. Name the three things that identify a file, with one example of each. (*Remember*)
3. Narrate the states a process passes through from "the user double-clicks the icon" to "the program is closed", using scheduled, descheduled and blocked at least once. (*Apply*)
4. Explain why closing a window does not always end a process, and how that relates to a slow machine. (*Understand*)

Summary

File management rests on three identifiers - name and extension, folder path, metadata - and the operating system's services for creating, moving and tracking files; a disciplined three-level scheme with date-first names turns a pile into a tree. Process management rests on the process, a program in execution with its address space, registers and open files; one processor runs one process at an instant, and the OS creates the illusion of many by rapid switching through the states running, ready and blocked. A slow machine is usually a crowded turn.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the three identifiers of a file and design a workable folder scheme.
- ☐ I can define a process and name its states with their transitions.
- ☐ I can explain a slow or "frozen" computer in terms of process management.

Try this at your practicum school

Ask the office for permission to look at the school's document store (or the desktop of the office computer). Sketch its current folder scheme and propose, on paper, a three-level replacement with a naming pattern. Show the proposal to the clerk and note one objection and one approval.

Summative task

Write a one-page "office rescue plan" for a school whose computer has two problems: no one can find last term's letters, and the machine slows every afternoon. Address the first with a file scheme (structure, naming pattern, where new files go) and the second with process management (what to check, what to close, what the states tell you).

Marking guide (10 marks):

- File scheme is workable: at most three levels, a naming pattern, and a rule for new files: 3 marks.
- The slowness diagnosis uses process states and the time-sharing idea rather than "it is old" (Analyse): 4 marks.
- Both problems connected to the OS functions from Topic 3.1: 2 marks.
- Clear, one-page plan: 1 mark.

Further reading

- Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2023). *Operating Systems: Three Easy Pieces* (version 1.10), chapter 4.
<https://pages.cs.wisc.edu/~remzi/OSTEP/>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapters 2 and 3. Open Textbook Library.
<https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Shelly, G. B., & Vermaat, M. E. *Discovering Computers*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)

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